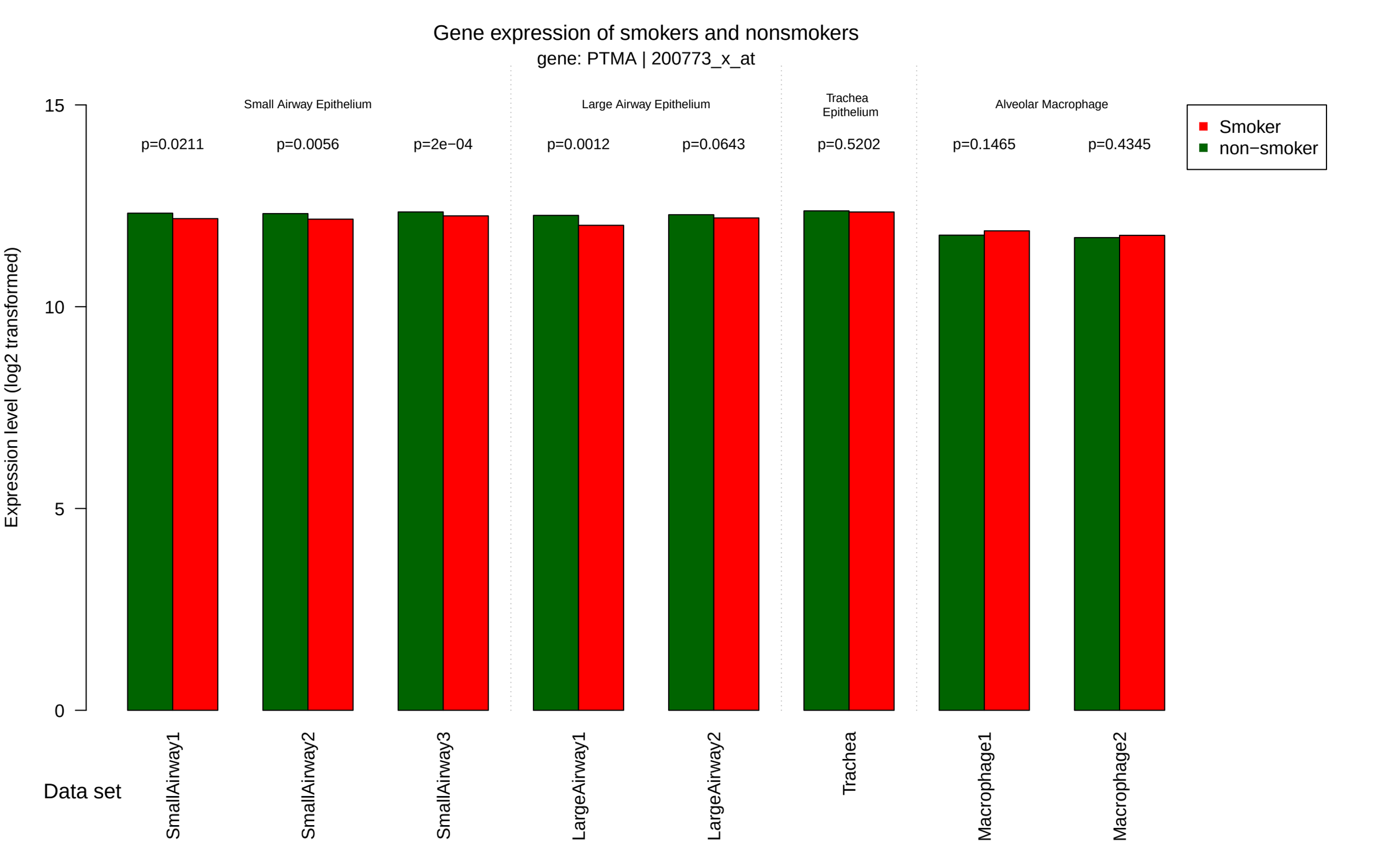
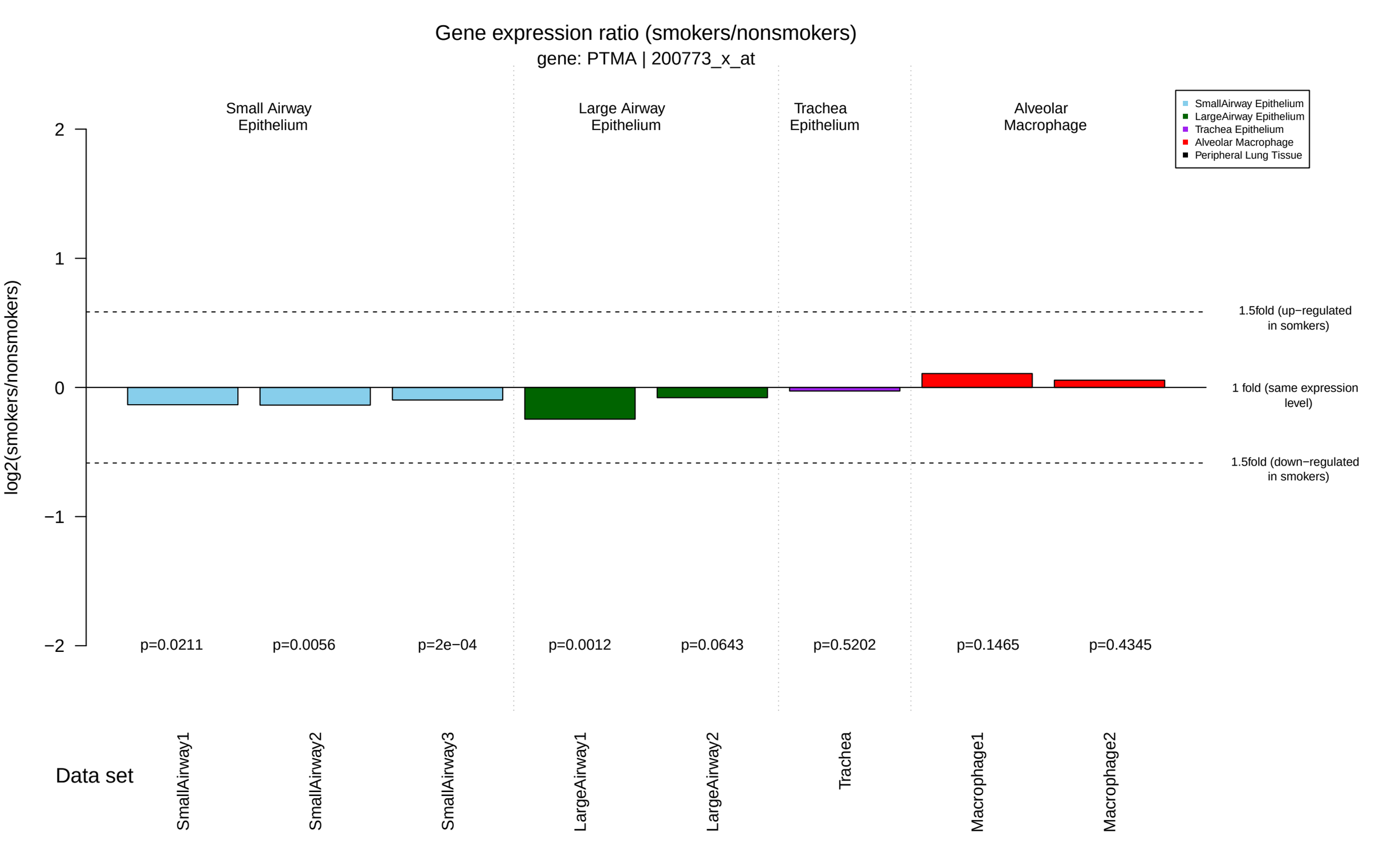
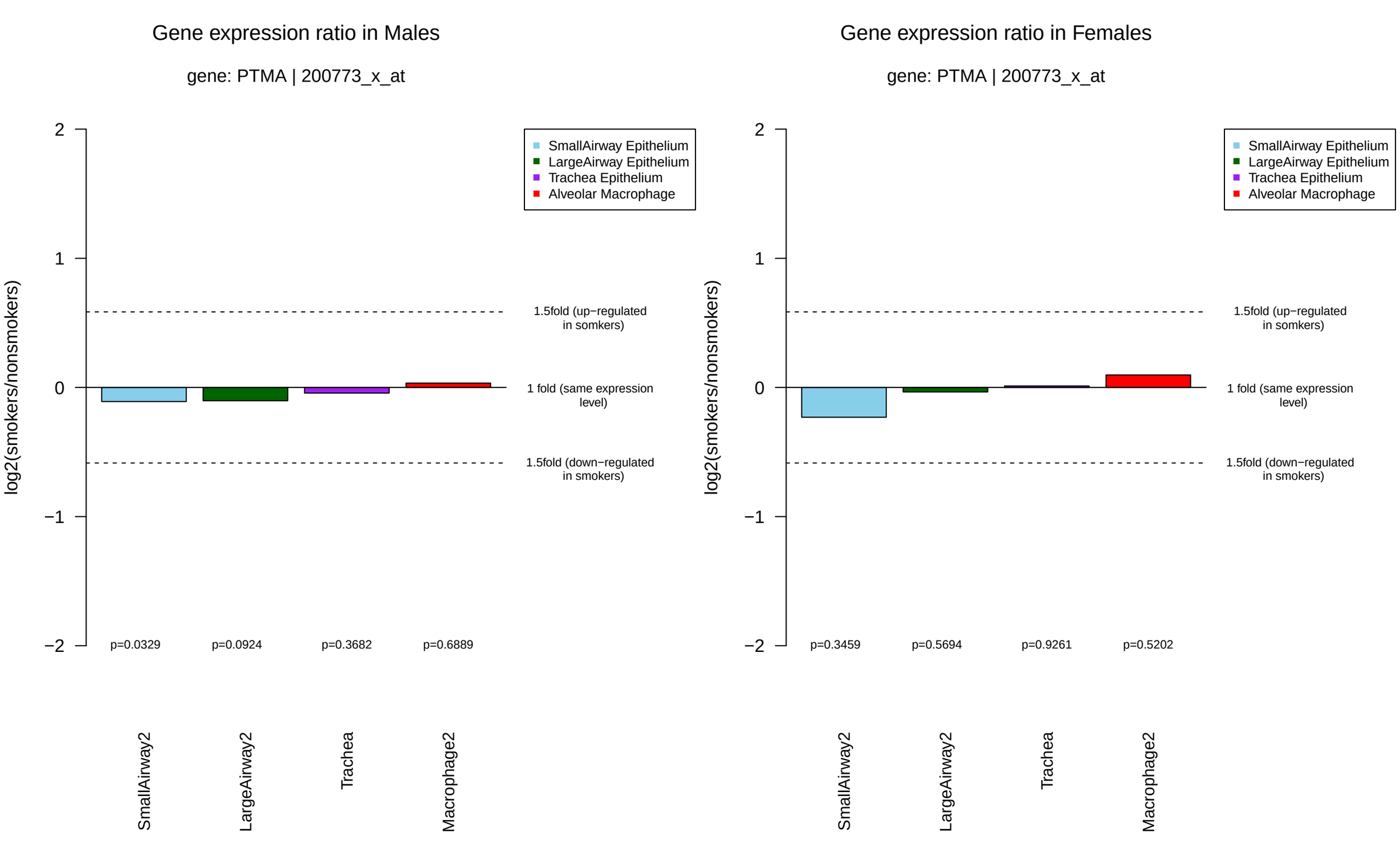
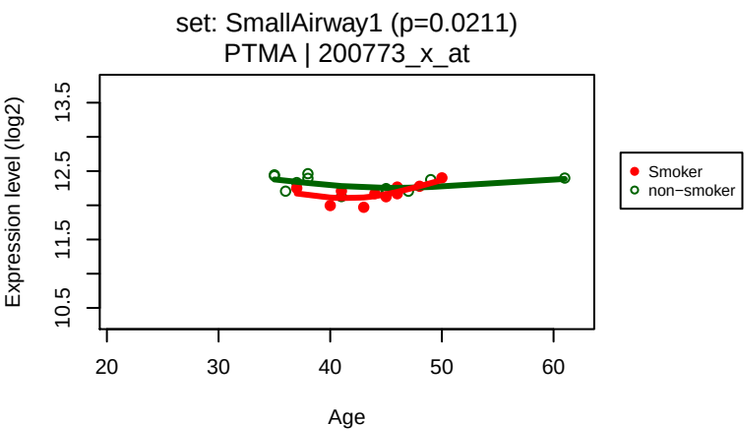
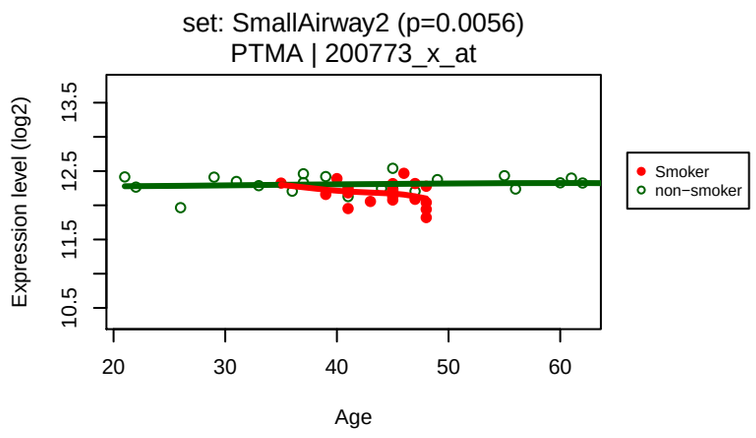
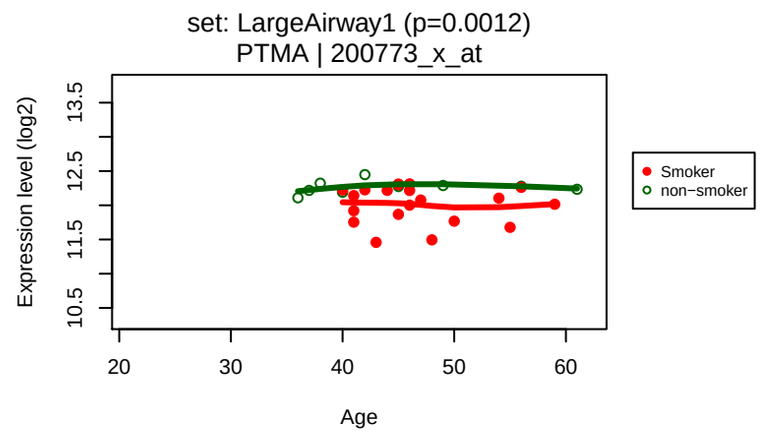

Gene expression of smokers and nonsmokers
gene: PTMA | 200773_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0211
p=0.0056
p=2e−04
p=0.0012
p=0.0643
p=0.5202
p=0.1465
p=0.4345
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

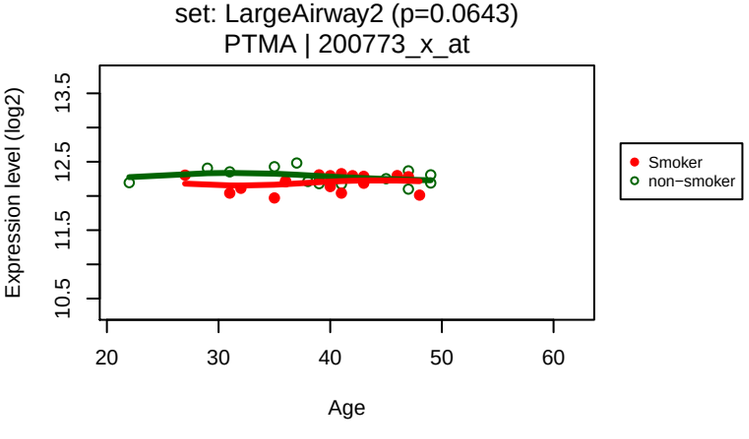

Gene expression ratio (smokers/nonsmokers)
gene: PTMA | 200773_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0211
p=0.0056
p=2e−04
p=0.0012
p=0.0643
p=0.5202
p=0.1465
p=0.4345
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

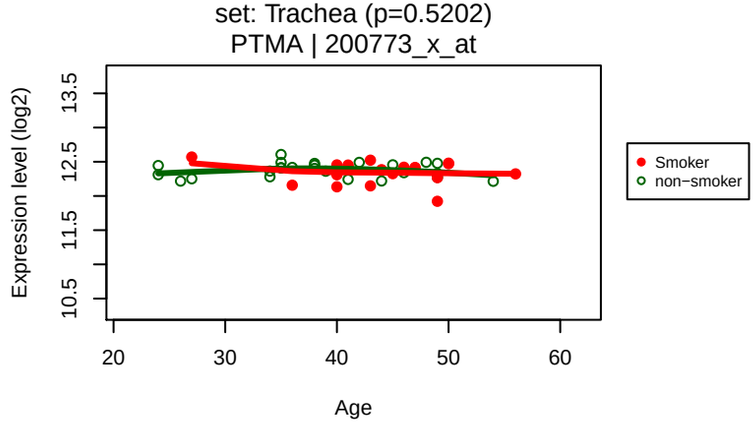

Gene expression ratio in Males
gene: PTMA | 200773_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0329
p=0.0924
p=0.3682
p=0.6889
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTMA | 200773_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3459
p=0.5694
p=0.9261
p=0.5202
SmallAirway2
LargeAirway2
Trachea
Macrophage2

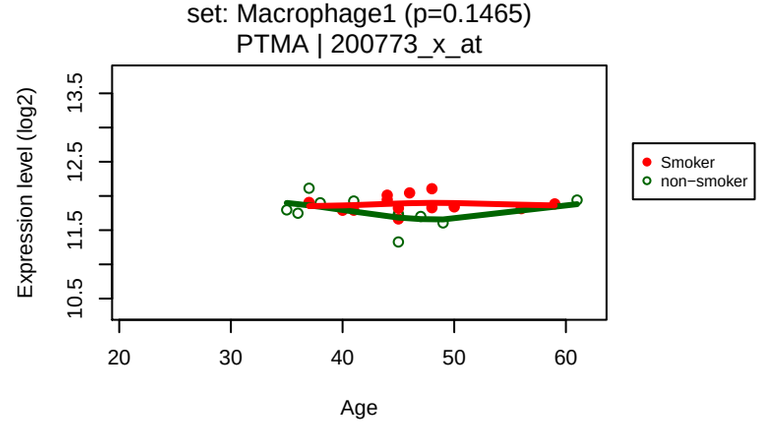

set: SmallAirway1 (p=0.0211)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker

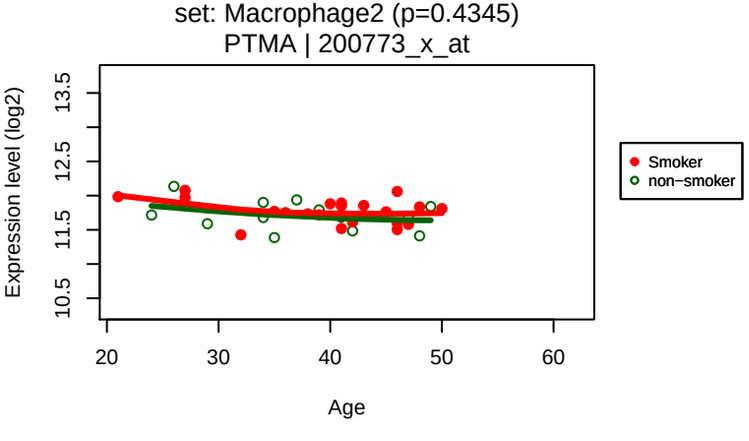

set: SmallAirway2 (p=0.0056)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker

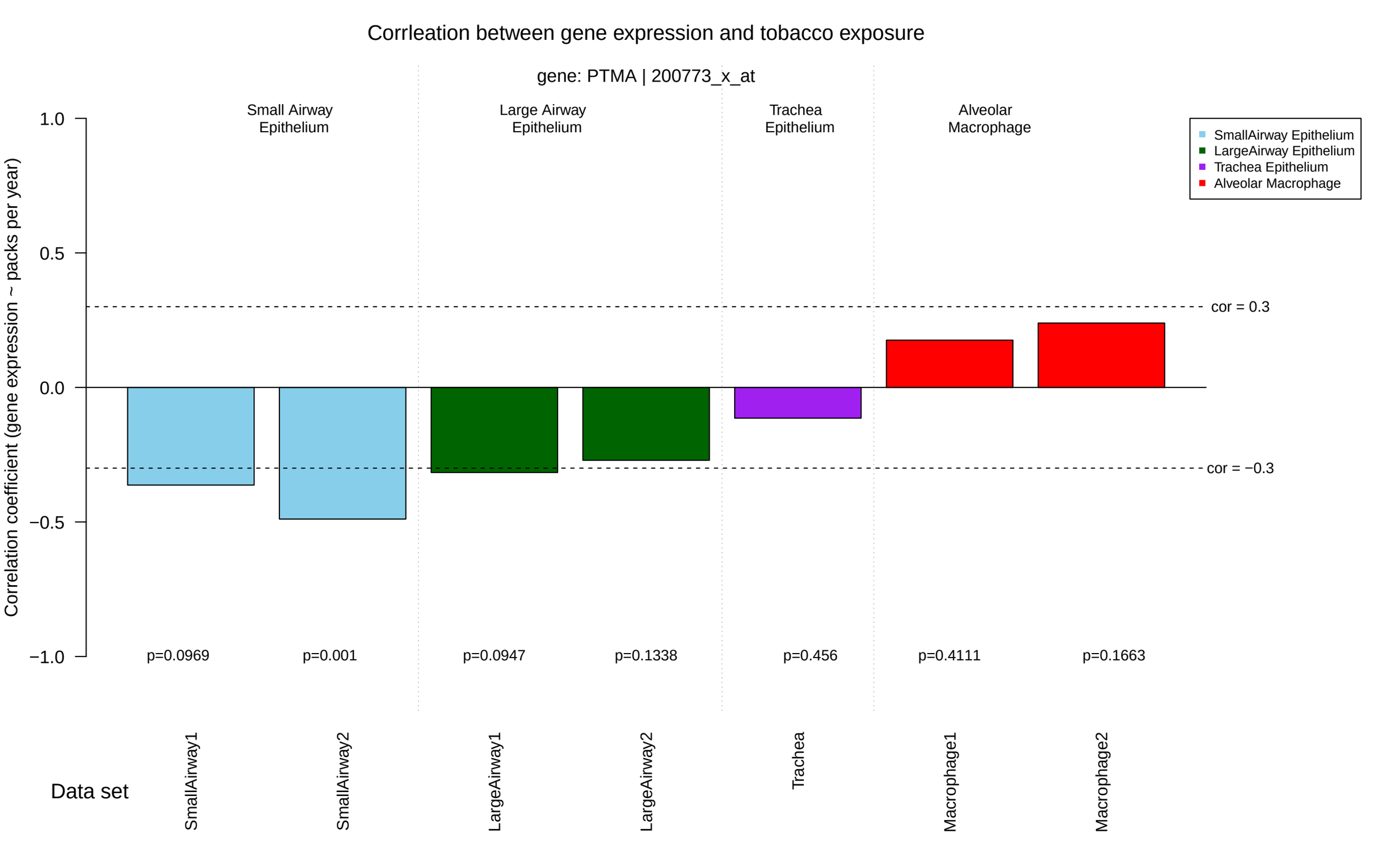

set: LargeAirway1 (p=0.0012)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0643)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5202)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1465)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4345)
PTMA | 200773_x_at
Expression level (log2)
10.5
11.5
12.5
13.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTMA | 200773_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0969
p=0.001
p=0.0947
p=0.1338
p=0.456
p=0.4111
p=0.1663
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2